

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Marks

Q.1 Solve the following.

- A) Explain the following string inbuilt functions with syntax and example: 4
i) len() ii) max () iii) min () iv) isdigit()
- B) Write a function that accepts two positive integers a and b and return a list of odd 4
numbers between a and b
- C) Write a python code to find the maximum of two numbers using function with 4
arguments.

Q.2 Solve any Two of the following.

- A) Explain the Importance of OOP. Explain the following concepts: 6
i) Class ii) Abstraction iii) Encapsulation
- B) Write python program to create a class Box, and calculate volume of a box, use 6
__init__ method
- C) What is exception handling? Write python code Program to handle multiple errors 6
with one except statement

Q.3 Solve Any Two of the following.

- A) What is linked list? Perform the polynomial addition using linked List 6
 $P(x) = 12x^4 + 2x^2 + 10$ $Q(x) = 9x^3 + 8x^2 + x$
- B) Write algorithm to evaluate postfix expression. evaluate the following postfix 6
expression using Stack: $6\ 4\ +\ 9\ 3\ -\ *$
- C) Define Queue. Implement queue using Python List. 6

Q.4 Solve Any Two of the following.

- A) Given input { 4371,1323,6173,4199,4344,9679,1989} and a hash function $h(X) = X$ 6
(mod 10). Show the resulting
1. Separate chaining table
2. Open addressing hash table using linear probing

3. Open addressing hash table using Quadratic probing

- B) Write algorithm for binary tree traversal techniques- Inorder, preorder and postorder 6
- C) Suppose that we have the following key values 7,16,49,82,5,31,6,2,44. 6
Sort the keys using max heap.

Q. 5 Solve Any Two of the following.

- A) Write python code to implement selection sort. 6
- B) Differentiate between Linear search and Binary Search method. Search key $x = 35$, $x = 28$ 6
using binary search and linear search method from following list: 10, 15, 25, 28, 35, 42, 55. Comment on number of comparisons required.
- C) Explain i) Greedy method ii) Big-O Notation 6

*** End ***